Bhaskar Akkena

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Education

Rochester Institute of Technology **May 2026**
Master of Science in Data Science, GPA: 3.6 *Rochester, NY*

Rochester Institute of Technology **May 2024**
Bachelor of Science in Computing and Information Technologies, GPA: 3.5, Dean's List *Rochester, NY*

Technical Skills

Languages: Java, Python, JavaScript, PHP, SQL, Bash
Databases Technologies: MySQL, MongoDB, Neo4j
Machine Learning: Pytorch, scikit-learn, Tensorflow, Keras, R, Pandas, Numpy, Seaborn, Plotly, Matplotlib
Cloud and Other Tools: AWS, Git, Docker, Kubernetes
Infrastructure: Windows Server, Linux, Active Directory, VMware
Certifications: Google Cloud Foundational Certification, AWS Cloud Practitioner

Work Experience

Rochester Institute of Technology **Aug 2022 – Dec 2022 — Jan 2025 – Present**
Teaching Assistant *Rochester, New York*

- Mentored over **150+ students** in **Task Automation and Analysis** using **Python** and **Bash**, enhancing students' scripting proficiency and automation efficiency by **30%**.
- Conducted **weekly lab sessions, code reviews, and grading** for **1000+ assignments**, delivering detailed feedback that improved student performance and engagement.
- Created supplemental exercises on Linux command-line operations, file handling, and cron jobs, increasing hands-on learning participation by **25%**.

Saunders College of Business **Jan 2023 – May 2024**
System Administrator *Rochester, New York*

- Managed and monitored **50+ enterprise systems** using **Active Directory**, **VMware ESXi**, and **Windows Server**, implementing **Zabbix** for real-time performance tracking and incident response, reducing downtime by **25%**.
- Engineered and executed the migration of applications from **Docker** to **Kubernetes**, designing namespaces, deployments, and ingress controllers to enhance scalability and deployment efficiency by **40%**.
- Led **hardware deployment and system refresh projects** with Lenovo vendors, provisioning **100+ systems** with **Ivanti integration**, imaging automation, and endpoint security hardening.
- Administered tier-2 **technical support** and **incident management** through **OTRS**, resolving **200+ tickets** with a **95% on-time resolution rate** and creating detailed system documentation for future reference.
- Collaborated with DevOps and Security teams to implement access auditing, **MFA enforcement**, and compliance checks aligned with **ISO 27001**, reducing unauthorized access incidents.

Projects

Evolutionary Search for Toxicity Control in Large Language Models (LLMs) **Aug 2025 - Present**

- Built an **automated evaluation pipeline** combining **text-based** (Perspective API, HateBERT) and **embedding-based toxicity metrics** to analyze over **5,000 LLM responses**.
- Designed a **context-aware scoring system** using **SentenceTransformers** to quantify semantic toxicity with **92% consistency**, and implemented an **evolutionary optimization loop** that reduced mean toxicity scores by **37%** through iterative refinement.
- Developed **Python-based logging and visualization modules** leveraging **JSON tracking** and **HuggingFace models** to benchmark model improvements.

Bias Characterization in AI-Generated Travel Narratives **Jan 2025 - Jun 2025**

- Designed a **comparative counterspeech framework** contrasting **GPT-4** and **IndicBERT** to interrogate **cultural bias and representational asymmetries** in multilingual travel discourse.
- Deployed an **end-to-end data pipeline** for comment ingestion, translation, prompt generation, and **automated toxicity–stance evaluation** across **500,000+ samples**.
- Developed a **context-sensitive scoring architecture** integrating **semantic embeddings** and **stance polarity metrics** to quantify latent bias expression.
- Performed **empirical bias mapping and visualization** using **embedding projections** and **Python-based analytics** to reveal discourse-level disparities between Western and Indic LLMs.

Fraudulent Job Posting Classification Model **Aug 2024 - Dec 2024**

- Investigated the problem of fake job postings in online job markets by analyzing publicly available records and worked to develop a **predictive model**, aiming to improve data reliability and protect job seekers from fraudulent opportunities.
- Engineered a robust data pipeline to perform **data cleaning**, and **preprocessing** on a dataset spanning 5 years of job postings across 50+ countries, with data on 100+ companies.
- Applied **text preprocessing**, **feature extraction**, and **model tuning** to analyze 10,000+ job postings, optimizing performance through 5-fold cross-validation to improve generalization and reduce overfitting.
- Implemented **Logistic Regression**, **Random Forest**, and **Gradient Descent** algorithms to predict fake job postings; achieved 72% F1-score, 80% Precision, and 67% Recall through **hyperparameter tuning**.